

MA3K7 Week 11 Rubric

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There are 3 different species of particle inside a contained box, labelled A , B and C . There are n number of species A , m number of species B and l number of species C . The particles move in straight lines, with their direction randomly determined at each collision. When a particle of species A collides with species B , they both change into 2 particles of species C . Similarly we get 2 particles of species B for a collision of A and C , and 2 particles of species A for a collision of B and C . Investigate what occurs for different values of n , m and l .

1 Entry

I KNOW

From the assumptions of the problem,

- There are 3 different species of particle.
- Each "game" has a fixed starting number for each species.
- If a game is to end at some point, it is required for it to only have one species remaining.
- We can consider each collision of 2 particles of different species to be a "round" in the game.
- A collision of 2 particles that are of the same species can be thought of in the same light as a collision of a particle with the wall.
- The total number of particles inside the box is always conserved. That is, it is unchanged by any number of collisions.

NOTATION

In this rubric I will refer to the following:

- A collection of successive rounds will be referred to as a game.
- We will refer to each iteration of the game as a round.
- If a game has a possible endpoint, we say that it converges. If such an endpoint does not exist, then the game does not converge or diverges.
- If a game converges to a species, we say this species is the winning species.
- We will use the notation N_A, N_B, N_C for the total number of particles of species A, B and C respectively.
- We will use N_S to denote the total number of an arbitrary species.
- Let $N = N_A + N_B + N_C$ denote the total number of particles over all species.
- Let $\varphi(N)$ denote the number of possible games for a total number of particles N .
- Let $\gamma(N)$ denote the number of possible convergent games for a total number of particles N .

I WANT

I will investigate what conditions are required for a game to converge. I want to find out whether or not there are certain values of N_A, N_B and N_C in which this can occur for. If there do exist certain values, perhaps we can deduce from a general trend a rule, and hence form a conjecture and prove it. This would get me closer to a potential numerical solution for the condition of convergence, giving me a possible clear and complete solution overall.

There is more to consider, however, than just possibilities of convergence. If we find certain games do converge, do some converge faster on average than others? Using the potential numerical rule for convergence, I aim to analyse various games with increasing particle counts to determine if I can accurately model their rate of convergence (measured by the average number of rounds).

After my investigative model, I wish to explore the combinatorial aspect of the problem. I want to see if for some N , there exists a formula for the total number of unique possible games. If this is calculated, I could then figure out a formula for the number of unique convergent games. Using this I would then know the fraction of games overall for any value of N that are convergent.

2 Attack

INTRODUCE

To approach the problem, I initially imagined it as a game with each turn representing a collision of 2 balls with each other. My original idea for the model also included the equal possibility of no balls colliding with each other, but rather just collide with the wall. However I quickly concluded that this feature would make run times much longer for the game, whilst without it we still have an accurate model for the situation. One could however argue that we would lose a little bit of "realism" of the actual situation, however the problem states that we do not need to worry about this.

I TRY

I began programming the game in Python, with multiple `if` statements for each condition. Then setting each condition to an integer, I used equal probabilities for each, and generated a random one for each round with `np.random.choice()`. To ensure I didn't get any negative probabilities, I made sure to add further `if` statements that checked whether a species had an N_S value that was greater than 0. The `game()` function took values for the starting N_S and a value for the number of rounds I wanted the game to run for. To make sure the game would have a stopping condition, further `if` statements were produced to check there were still at least 2 positive values of N_S . If not, the loop would break and the game could end. From here I was now ready to start making observations for different values of N_A, N_B and N_C .

I FIND

After trialing lots of different games, I do indeed find there exists some games that do converge to a single species at a certain point, provided the round size is sufficiently large enough. I

AHA!

notice, for example, that this always occurs for whenever $N_A = N_B = N_C$. However in this situation, the species that wins is not unique. But, if I change exactly one of the N_S to 1 or 2 values fewer or 1 or 2 values higher, this species now always wins. However, once this species is exactly 3 values fewer or higher, any of the 3 species can win again. This discovery lead me to believe there was some form of relationship between successful convergence and $n \pmod{3}$.

CONJECTURE

JUSTIFY

This relationship is indeed noteworthy. If for example we had a collision of species A with species B , we would find that $N_A \rightarrow N_A - 1$, $N_B \rightarrow N_B - 1$ and $N_C \rightarrow N_C + 2$. Observing that $(N_A - 1) - (N_B - 1) = N_A - N_B$, we note that collisions of two particles do not affect the difference in the total number of particles in the species itself. This means if we had $N_A = N_B = n$ number of particles, they could all be theoretically paired up to eventually converge to species C . This situation is, however, only possible if for $N_a = n$ and $N_B = m$ with $m \geq n$ if we have $m \equiv n \pmod{3}$ ¹. This is as for each collision of B with C , $N_A \rightarrow N_A + 2$ whilst $N_B \rightarrow N_B - 1$, making the difference between N_A and N_B change by -3 . Thus theoretically after successive collisions between B and C , we could reach a point where $N_A = N_B$ and hence find convergence

¹Note: This still holds if instead we had $m \leq n$.

overall. See **Claim 1** in the **Solution** section for a full, formal proof of this.

I WANT I had now successfully found a concrete mathematical structure behind the way the particles would or would not converge. But I still needed to research whether or not there existed a model one could use to figure out the mean number of rounds it would take for some N number of particles to converge to some species.

I TRY I needed to now check that the mean round in which a convergent game actually converged was consistent. I created a function that would take multiple samples of the same game a fixed number of times and then calculate its mean. The function repeated this process another set number of times, and then placed its findings into a `numpy` array. It returned the array itself

I FIND and the value of the mean of all the means. I tested this with various convergent games, all differing in values of N , and discovered that there was at least a good level of consistency in the lower values of N . With larger N , however, it appeared the games were not as consistent. I suspected at this point that this was probably due to an increase in variance of the convergent rounds. See Fig. 1 for an example of this with the game where $N_A = 8$, $N_B = 10$ and $N_C = 11$.

```
(array([1624.001, 1662.875, 1609.751, 1686.98 , 1618.313, 1769.219,
        1656.038, 1696.775, 1765.325, 1693.181, 1621.04 , 1595.045,
        1710.194, 1636.64 , 1600.994, 1680.68 , 1557.854, 1647.473,
        1721.417, 1614.32 , 1670.333, 1718.243, 1690.661, 1590.416,
        1563.119, 1623.2 , 1650.461, 1663.1 , 1678.493, 1506.113,
        1724.312, 1629.251, 1595.639, 1597.919, 1655.132, 1678.373,
        1704.563, 1638.224, 1662.227, 1626.131, 1697.81 , 1656.8 ,
        1664.378, 1564.319, 1664.852, 1637.831, 1583.675, 1603.079,
        1618.988, 1683.389, 1688.264, 1581.089, 1693.106, 1587.893,
        1694.906, 1582.463, 1713.593, 1652.321, 1599.92 , 1708.688,
        1757.456, 1639.412, 1704.803, 1684.439, 1742.897, 1624.364,
        1653.617, 1677.422, 1647.299, 1673.783, 1623.95 , 1707.749,
        1649.534, 1680.395, 1685.252, 1649.795, 1642.58 , 1742.438,
        1711.046, 1740.545, 1695.134, 1670.09 , 1633.841, 1698.404,
        1738.85 , 1623.794, 1578.527, 1653.458, 1665.338, 1754.984,
        1608.344, 1631.855, 1644.743, 1764.524, 1659.764, 1632.644,
        1649.366, 1687.508, 1654.733, 1636.184])), 1658.3225000000002)
```

Figure 1: An example output array for the mean round number of a game

I TRY Now I had found there was an element of consistency among the means of multiple games, I wanted to see if I could create a model that could predict these means. I plotted 100 convergent games on a scatter graph, and used `curve_fit` from `scipy.optimize` with a quadratic function to see if the model accurately represented a trend. The games began with $N = 7$, with the N_C value increasing by 3 each time and the N_A and N_B values being fixed at 2 and 5 respectively.

JUSTIFY This was done so we can ensure every unique game is in fact definitely convergent, as otherwise the data would be filled with outliers that always shoot to the full 90,000 round count cap.

I FIND I found that for lower values of N , the model was a good representation of the data. Whilst for higher values of N , this became less true, with the data looking more scattered. Due to loading times being still quite long, I needed to keep the number of trials per unique game to a minimum. It is worth noting that this is true for Fig. 2, where only 100 trials of each game had been used. The problem was that this aspect to the problem was entirely random how many times it would take to run a simulation. We can justify this by a simple looping logic. For every collision between an A particle and a B particle, the following 2 collisions could be between a

STUCK

B and a C particle and an A and a C particle. This means that $N_A \rightarrow N_A - 1 + 2 - 1 = N_A$, $N_B \rightarrow N_B - 1 + 2 - 1 = N_B$ and $N_C \rightarrow N_C - 1 + 2 - 1 = N_C$, leading us back to the exact same situation as we were in previous to this. This looping possibility could happen an infinite number of times, meaning the round end could differ massively. It was thus clear that a faster game function would need to be created to manage a larger sample size for each unique game.

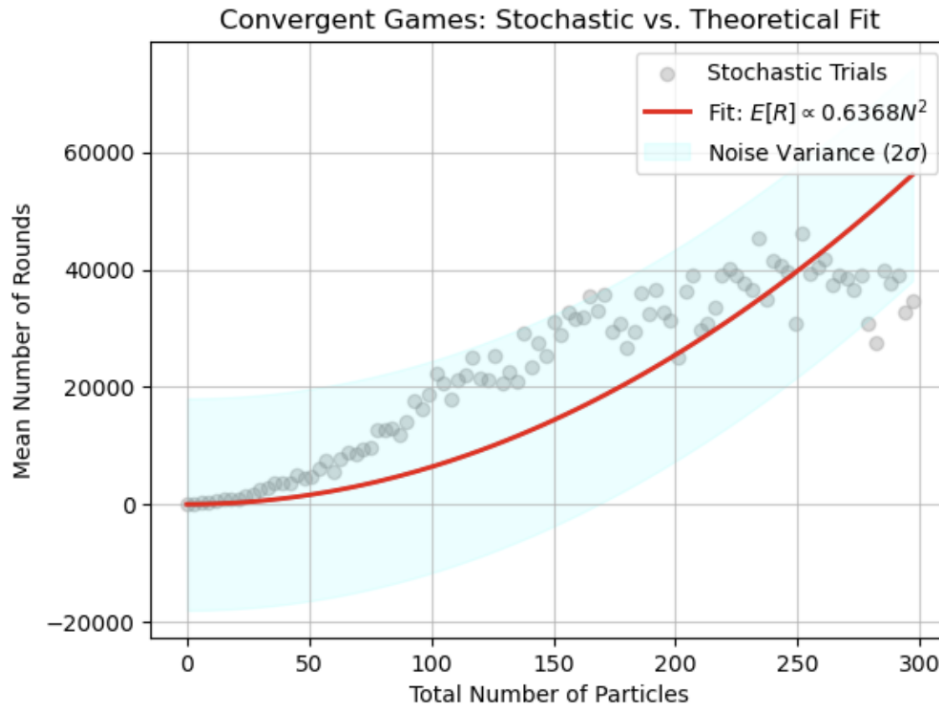


Figure 2: Stochastic trials versus theoretical fit for convergent games of increasing N value.

I TRY

To run multiple games, the original code I had written was far too slow. Thus I began coding a new and faster function for the game, `game2()`. This function sped the code up by utilising the `random.choice()` function from the `random` library. This is much faster than the `numpy` function for single values of randomly selected data. The initial function also used a separate checking function, which had to be called back for every iterative round. In the new function this was implemented into the function itself. Once everything had been successfully coded, I ran a tester of both functions for a theoretical 1000 round simulation, to compare which one was actually faster. The `game2()` function was much quicker, by a whole unit of measurement. Now I could map multiple games

JUSTIFY

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Since each collision represents a stochastic choice between 3 options, we can think of the game as acting as a random walk on a 2 dimensional triangular lattice. Within stochastic processes, the expected duration of a converging situation acts scales quadratically. Thus, theoretically a quadratic model is an accurate representation for the data. The light blue shaded area on the plot represents the noise variance of 2 standard deviations above and below the fitted curve line.

I FIND

All the scattered data managed to fit in-between the shaded blue area. However, the data slightly increases upwards above the red fitted line, potentially suggesting a quadratic fit may not be the most accurate model for the data. It is worth noting the increased noise variance as the total number of particles in the system also increased. This is often a key feature of random walks, and if anything strengthens the validity of the model. But rather than spending

JUSTIFY

time justifying my model through simple observations, I decided it was necessary to perform a formal statistical analysis on the data.

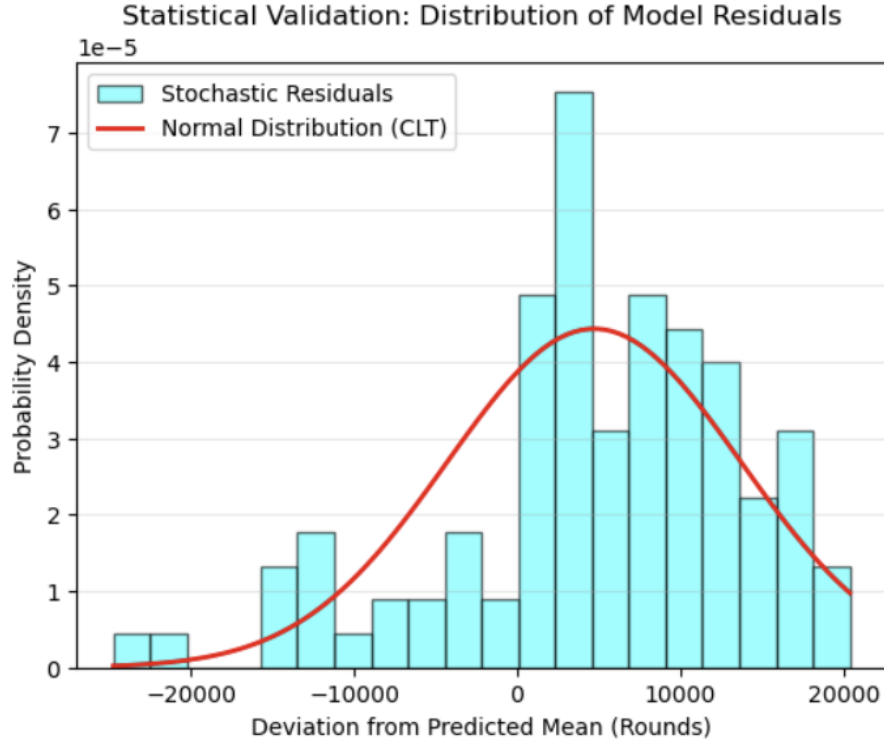


Figure 3: Central Limit Theorem analysis of stochastic data with quadratic model.

Fig. 3 displays a clear empirical validation for the usage of a quadratic relationship to model the data. There is a clear alignment from the stochastic findings with the theoretical fitted normal distribution curve, which shows the model is unbiased. Most importantly, the agreement to the Central Limit Theorem presents a clear explanation for the deviations from the predicted mean. It shows these deviations are simply a reflection of stochastic noise, and hence do not invalidate the model. Thus, I conclude my quadratic model is an accurate representation of the scenario.

I WANT

The last thing to investigate was the number of potential game situations of particles that can be formed. Additionally from this, I wanted to investigate the number of possible convergent games. From there I would know for any value of N what fraction of its possible games are convergent.

I TRY
JUSTIFY

I calculated the function on Python, using loops to go through every single possible combination. To understand this, we can imagine the situation for 2 species instead of 3. Here the possible number of unique games would be $N + 1$, due to the symmetry of the possibilities (we note here the distinction of the possibility of 0). From here we need only iterate each $a \in \{i \in \mathbb{N} : 0 \leq i \leq N\}$ the $N + 1 - a$ possibilities and count them up.

AHA!
CONJECTURE

I found , after iterating the first 10 possibilities of N , that this mimicked the triangle numbers. I thus hypothesised that the formula for the possible games of N particles was the formula for the $(N + 1)$ th triangle number. I.e.

$$\varphi(N) = \frac{(N + 2)(N + 1)}{2}.$$

I FIND

Upon research, I found a theorem that directly explained this scenario mathematically: *The Stars and Bars Theorem*. This theorem states that the number of possible ways to sort n stars into k bins is

$$\binom{n+k-1}{k-1} = \frac{(n+k-1)!}{n!(k-1)!}.$$

JUSTIFY

The theorem relates to our scenario, as for our possible games, we have N particles (our stars) to fit into 3 bins (our species of stars). Thus we have a matching result with our conjecture by applying the theorem for $n = N$ and $k = 3$:

$$\binom{n+k-1}{k-1} = \binom{N+3-1}{3-1} = \binom{N+2}{2} = \frac{(N+2)(N+1)}{2}.$$

I TRY

We now just needed a way to calculate the number of possible convergent games for any value of N . My strategy was to find a formula for the non-convergent games. For this, I focused on the third property as specified in the **Solution** section. As per this, we require all the N_S to be distinctive modulo 3. I set $N_A = 3i$, $N_B = 3j + 1$ and $N_C = 3k + 2$ for some i, j, k .

I FIND

From this, I observed that the sum of these values was a multiple of 3, meaning $N = 3m$ for some m . After some algebraic manipulation, usage of *The Stars and Bars Theorem* and some substitution, I concluded that the formula for the number of convergent games for some N is

$$\gamma(N) = \varphi(N) - \frac{N^2 + 3N}{3} = \frac{(N+2)(N+1)}{2} - \frac{N^2 + 3N}{3}$$

See **Claim 4** in the **Solution** section for a full proof of this. We can combine these two formulas to find the fraction of convergent games of total games for any value of N :

$$\begin{aligned} \frac{\gamma(N)}{\varphi(N)} &= \left(\frac{(N+2)(N+1)}{2} - \frac{N^2 + 3N}{3} \right) \bigg/ \left(\frac{(N+2)(N+1)}{2} \right) \\ &= 1 - \left(\frac{2N(N+3)}{3(N+2)(N+1)} \right). \end{aligned}$$

I TRY

Calculating a limit for this, I found the limit of $\gamma(N)/\varphi(N)$ as $N \rightarrow \infty$ was $1/3$. Finally to investigate how quickly this function did converge, I tested the values of $\gamma(N)/\varphi(N)$ for $N = 3, 4, 5, 6, 7, 8$. Their values were $2/5, 17/45, 23/63, 5/14, 19/54, 47/135$ respectfully. I concluded that all these values of $\gamma(N)/\varphi(N)$ are all close to the limit of $1/3$, with particular focus to the sudden closeness of $23/63$ for $N = 5$. I conclude that this fraction converges at a fast rate.

[2/5, 17/45, 23/63, 5/14, 19/54, 47/135, 19/55, 34/99, 40/117, 31/91, 107/315, 61/180, 23/68, 155/459, 173/513, 32/95, 106/315, 233/693, 85/253, 139/414, 151/450, 109/325, 353/1053, 190/567, 68/203, 437/1305, 467/1395, 83/248, 265/792, 563/1683, 199/595, 316/945, 334/999, 235/703, 743/2223, 391/1170, 137/410, 863/2583, 905/2709, 158/473, 496/1485, 1037/3105, 361/1081, 565/1692, 589/1764, 409/1225, 1277/3825, 664/1989, 230/689, 1433/4293, 1487/4455, 257/770, 799/2394, 1655/4959, 571/1711, 886/2655, 916/2745, 631/1891, 1955/5859, 1009/3024, 347/1040, 2147/6435, 2213/6633, 380/1139, 1174/3519, 2417/7245, 829/2485, 1279/3834, 1315/3942, 901/2701, 2777/8325, 1426/4275, 488/1463, 3005/9009, 3083/9243, 527/1580, 1621/4860, 3323/9963, 1135/3403, 1744/5229, 1786/5355, 1219/3655, 3743/11223, 1915/5742, 653/1958, 4007/12015, 4097/12285, 698/2093, 2140/6417, 4373/13113, 1489/4465, 2281/6840, 2329/6984, 1585/4753, 4853/14553, 2476/7425, 842/2525, 5153/15453, 5255/15759, 893/2678]

Figure 4: Array of $\gamma(N)/\varphi(N)$ for increasing, stepwise value of N .

3 Solution

If for a starting state (N_A, N_B, N_C) we have $N_A \equiv N_B \equiv N_C \pmod{3}$, then the game converges to any of the 3 species. If instead we had only $N_A \equiv N_B \pmod{3}$ with $N_A, N_B \not\equiv N_C \pmod{3}$, then the game converges to species C . If $N_A \not\equiv N_B \not\equiv N_C \pmod{3}$, then the game does not converge to any species. The mean number of rounds it takes for a unique convergent game to end is stochastic in nature, and can be modelled using the relationship $\mathbb{E}[R] \propto 0.6368N^2$. The game can be thought of as a random walk, and thus this model matches the theoretical model for the system. The formula for the number of unique games for N total particles is $\varphi(N) = (N+2)(N+1)/2$. Out of these games we also know the formulas for the number of convergent ($\gamma(N)$) and non-convergent games. The formula for convergent games over total games, $\varphi(N)/\gamma(N)$, for some N particles converges to $1/3$ as $N \rightarrow \infty$.

Claim 1. *For a game with fixed values N_A , N_B and N_C , the following properties apply:*

1. *If $N_A \equiv N_B \equiv N_C \pmod{3}$, then the game converges to any of the 3 species.*
2. *If $N_A \equiv N_B \pmod{3}$, then the game converges to species C .*
3. *If $N_A \not\equiv N_B \not\equiv N_C \pmod{3}$, then the game does not converge to any species.*

Proof. Consider the situation where a particle of species A collides with a particle of species B . Let us describe the initial state of all N_S by the triplet (N_A, N_B, N_C) . After the collision, our triplet becomes the state $(N_A - 1, N_B - 1, N_C + 2)$. We observe the differences between the N_S after this state

$$(N_A - 1) - (N_B - 1) = N_A - N_B \quad (N_A - 1) - (N_C + 2) = N_A - N_C - 3.$$

For the first, we observe that difference between these two N_S remains constant after the collision. For the second, we note that the difference between these two is invariant under $\pmod{3}$. Thus we find the difference between any two N_S after a collision is invariant under modulo 3. Thus we find for all species S_1 and S_2 and some constant integer m that

$$N_{S_1} - N_{S_2} \equiv m \pmod{3}.$$

Property 2: For a game to converge to species C , we require the game to end in the state $(0, 0, N)$. In this final state we have $N_A - N_B = 0 - 0 = 0$ which implies $N_A \equiv N_B \pmod{3}$. If we have in the initial state that $N_A \equiv N_B \pmod{3}$, then we must have $N_A - N_B \equiv 0 \pmod{3}$. But as this value is invariant, we can therefore reach a state wherein $N_A - N_B = 0$, and thus game converges to C .

Property 1: If we have $N_A \equiv N_B \equiv N_C \pmod{3}$, then the difference between any 2 species is congruent to 0 $\pmod{3}$. I.e. we have that

- $N_A - N_B \equiv 0 \pmod{3}$.
- $N_B - N_C \equiv 0 \pmod{3}$.
- $N_A - N_C \equiv 0 \pmod{3}$.

²Similarly for $N_B \equiv N_C \pmod{3}$ and $N_A \equiv N_C \pmod{3}$.

Since any pairwise difference is congruent to 0 (mod 3), it is possible to reach any of the possible final convergence states $(N, 0, 0)$, $(0, N, 0)$ and $(0, 0, N)$. Thus the game can converge to any species.

Property 3: If the initial states N_A , N_B and N_C are all congruent to distinct modulo 3, then they are congruent to a permutation of $\{1, 2, 3\} \pmod{3}$. Calculating the differences, we find that

- $N_A - N_B \equiv 1 \text{ or } 2 \pmod{3}$.
- $N_B - N_C \equiv 1 \text{ or } 2 \pmod{3}$.
- $N_A - N_C \equiv 1 \text{ or } 2 \pmod{3}$.

For a game to converge to any species, the difference of the other N_S must equal 0 in the final state and hence must also be congruent to 0 (mod 3) in the final state. Due to the invariant property, however, the difference at the start must also be congruent to 0 (mod 3). We see clearly here for any of the pairwise relations that this does not occur, and so therefore the game does not converge to any species. $\square$

Claim 2 (Stars and Bars Theorem). *For n items being sorted into k bins, the number of possible ways to do this is*

$$\binom{n+k-1}{k-1} = \frac{(n+k-1)!}{n!(k-1)!}.$$

Proof. Consider a sequence of n stars (denoted $\star$) and $k-1$ bars (denoted $|$). For example if $n = 8$ and $k = 4$, we could have the sequence $\star\star|\star|\star\star|\star\star$. The bars act as dividers for the n stars to be sorted into k bins. Thus for each unique sorting of n stars into k bins, there exists a unique "star and bar sequence" that represents it. Therefore we can define a bijection from the set of possible bin sorting options to the set of possible star and bar sequences.

If we consider each star and bar sequence as a string, the length L of the string must be equal to the sum of n stars and $k-1$ bars. To create a unique sequence, we need only choose which of the L positions along the string will be occupied by the $k-1$ bars. This is exactly L choose $k-1$, and so by the definition of the choose formula

$$\binom{L}{k-1} = \binom{n+k-1}{k-1} = \frac{(n+k-1)!}{(k-1)![n+k-1-(k-1)]!} = \frac{(n+k-1)!}{n!(k-1)!}.$$

$\square$

It is worth noting that we can also generate a unique star and bar sequence by choosing n of the L positions are to be occupied by stars. The formula still holds in this case, due to the symmetry of the choose function.

Claim 3. *For 3 distinctive species of particle, the number of unique possibilities of games for N total particles is*

$$\binom{N+2}{2} = \frac{(N+2)(N+1)}{2}.$$

Proof. As explained before in the **Attack** section of the rubric, the case for our situation is exactly *Stars and Bars* for $n = N$ and $k = 3$. Thus we have

$$\binom{n+k-1}{k-1} = \binom{N+3-1}{3-1} = \binom{N+2}{2} = \frac{(N+2)!}{N!(2!)}$$

which simplifies to

$$\frac{(N+2)!}{N!(2!)} = \frac{(N+2)(N+1)N!}{(2 \cdot 1)N!} = \frac{(N+2)(N+1)}{2}.$$

□

Claim 4. For 3 distinctive species of particle, the number of unique possibilities for convergent games of N total particles is

$$\gamma(N) = \varphi(N) - \frac{N^2 - 3N}{3} = \frac{(N+2)(N+1)}{2} - \frac{N^2 + 3N}{3}.$$

Proof. We attempt to find a formula for M number of possible non-convergent games. We note for this to occur, the N_S must all be distinctive modulo 3. Thus we let $N_A = 3i$, $N_B = 3j + 1$ and $N_C = 3k + 2$ for some i, j, k . Summing these we get

$$N_A + N_B + N_C = 3i + (3j + 1) + (3k + 2) = 3(i + j + k + 1).$$

As these sum to a multiple of 3, we can set $N = 3m$ for some positive integer m . So we now have the relationship $i + j + k = m - 1$. This is the exact situation to apply **Claim 2**, giving the number of possible ways to organise $m - 1$ games into i, j , and k bins being

$$\binom{(m-1) + 3 - 1}{3 - 1} = \binom{m+1}{2}.$$

We can organise the residues, $\{0, 1, 2\}$, into the species $\{A, B, C\}$ in $3! = 6$ different ways. Thus to finally obtain a value for M , we multiply our former calculation by 6, giving us

$$M = 6 \cdot \binom{m+1}{2} = 6 \cdot \frac{m(m+1)}{2} = 3m(m+1).$$

Substituting $m = N/3$ back into the equation,

$$M = 3 \left(\frac{N}{3} \right) \left(\frac{N}{3} + 1 \right) = N \left(\frac{N+3}{3} \right) = \frac{N^2 + 3N}{3}.$$

As this is the value for the number of non-convergent games, to find the number of convergent games we need only to find the difference between this and the the total number games. Therefore

$$\gamma(N) = \varphi(N) - M = \frac{(N+2)(N+1)}{2} - \frac{N^2 + 3N}{3}.$$

□

Claim 5. The limit of the function $\gamma(N)/\varphi(N)$ is

$$\lim_{N \rightarrow \infty} \left(\frac{\gamma(N)}{\varphi(N)} \right) = \lim_{N \rightarrow \infty} \left[1 - \left(\frac{2N(N+3)}{3(N+2)(N+1)} \right) \right] = \frac{1}{3}$$

Proof. We can simplify the expression for $\gamma(N)/\varphi(N)$ to

$$\frac{\gamma(N)}{\varphi(N)} = 1 - \left(\frac{2N^2 + 6N}{3N^2 + 9N + 6} \right).$$

The limit of the latter expression is determined only by its highest ordered term, i.e. the N^2 term. This gives us

$$\lim_{N \rightarrow \infty} \left(\frac{2N^2 + 6N}{3N^2 + 9N + 6} \right) = \frac{2}{3}.$$

Thus by the algebra of limits,

$$\lim_{N \rightarrow \infty} \left(\frac{\gamma(N)}{\varphi(N)} \right) = \lim_{N \rightarrow \infty} (1) - \frac{2}{3} = 1 - \frac{2}{3} = \frac{1}{3}.$$

□

4 Review

CHECK

I believe I have thoroughly justified and double checked my work and solutions throughout. For every analytical feature of the problem, I have both explored well in Python and properly proven in the **Solution** section. The problem had stochastic elements to it, which I felt I handled in a way that was both productive and thorough. I provided statistical testing for the fit of my data with my model, and produced theoretical reasoning behind my quadratic fit. I feel my solution is complete overall, balancing many different methodologies to analyse the problem.

REFLECT

Although the problem does not ask us to consider a physically realistic scenario, for the sake of reflection we should do so. The first thing to note is the affect of friction and other forces that the particles would experience upon interaction. In a genuine, realistic scenario, these would affect the speeds in which the particles move, and hence the probabilities in the following interaction would not be the same as previous. Our model also does not account for interactions the particles may have with the walls. This could mean that the ending round for a convergent game could in fact be a lot higher than what was observed in our data. I do not think, though, that this would massively affect the data that has been collected. The more number theoretic and combinatorial aspects to the problem would not be changed at all with this introduction. In terms of the stochastic data that has been collected, this has more potential to change. But I predict the general trend observed would still be the same.

I found the use of Python in this assignment very useful. Firstly, it was a crucial tool to model the situation and spot the initial patterns that meant I could prove certain invariant properties. It provided a good verification for the formula of $\varphi(N)$, and served as a great guide for me when the triangle numbers popped out upon the creation of my iterative function. Without seeing this, it would have been difficult to know the direction that I was meant to head in. Most importantly, however, Python was essential in the stochastic process and quadratic curve fitting part of the investigation. I used Python successfully to create an accurate model (to the best of my ability) and test its validity through statistical testing. There has not been a single assignment so far that has not heavily relied upon the usage of Python, and this project was by far no different.

I have expanded my use of the *I Want* introduction, located towards the end of the **Entry** section. In previous rubrics this was always marked as too vague and short, whilst in this rubric I feel I have given the section a clear plan and direction for the project. Naturally, this is to be expected for the project, as the final solution is required to be more complete and thorough than a standard assignment solution. Hence, the need for a thorough plan is much more critical, but nevertheless I feel I made a significant improvement in this area.

Throughout this module my theorem proving and Python skills have greatly improved. This rubric is stuffed with multiple proofs, and thoroughly backed up discoveries in Python. I am confident in my ability to successfully write a mathematical rubric, and feel pleased with my gradual improvement of this overall over time.

If I were to do something else differently in this project, I would have potentially made an even faster game function in Python. The final game function that I used was really quite slow and I feel the plots that I had created could have benefitted from a higher sample size. Perhaps if I had done this, there could have been less noise in Fig. 2, and I could have obtained an even more accurate model for the situation. That being said, the purpose of obtaining the model was never to find a hyper-realistic and accurate line, but rather to prove the model was indeed quadratic in nature.

EXTENSION

An interesting thought experiment for this problem could be to imagine the 3 species of particle as rock, paper or scissors. Then just like in the traditional game, rock would "beat" scissors, paper would "beat" rock and scissors would "beat" paper. In this scenario the changes to N_R , N_P and N_S would be negative after each interaction. For example in an interaction with a rock and a paper the changes would leave both N_S and N_P unchanged and $N_R \rightarrow N_R - 1$. Here there would be no such thing as a convergent or non-convergent game and the number of rounds for each game would be consistently the exact same.

It could be interesting to see what would happen if the problem was made into a real game played with actual people being an individual particle. The goal would be to choose the team at the beginning that ends up with the most players at the end. The most obvious tactic for this game, would be to wait until every other player had chosen their team before picking, so that you could attempt to pick the team that had a distinctive modulo 3 number of players, when compared with the others. In practice, however, this would be difficult to successfully replicate. Unless the playerbase per game was very small, it would be difficult to know exactly when the last person had actually picked their team, leaving you to inaccurately choose your team. Even if you had successfully navigated the team selection, there is a high likelihood that each game would have a time limit (similar to our simulation's round limit), which would simply mean the team with the highest number of people at the start was most likely to win.

Let us now consider the case where we have k different species of particle. This suddenly becomes very different, as the invariant properties are not so easy to consider. For example if a collision were to occur between species A and B in $k = 4$, the total change of the system would not be congruent to 0 (mod 4) as required. Thus it is harder to find a universal invariant property. This is, however, dissimilar to the case where k is prime.

In the case where k is prime, one could devise a weighted sum, $W = \sum_{j=1}^k j \cdot N_j$. From here, one can often show that $W \pmod{k}$ is an invariant, and hence could devise a system of potential rules for convergent games.

Although I will not be performing any more statistical tests or large scale Python plots, one can still easily use a similar quadratic model to the model that was used in the case for $k = 3$. This is as no matter how high the species count is, the problem still acts as a random walk. Thus, the expectation still remains quadratic and the model for the system holds.

Despite the limitations of the other aspects to this model, it is still easy to calculate the number of possible unique combinations for games. This is as *The Star and Bar Theorem* holds for any value of k bins. We thus conclude the total number of possible games, φ_k for some N total number of particles is

$$\varphi_k(N) = \binom{N+k-1}{k-1} = \frac{(N+k-1)!}{N!(k-1)!} = \frac{\prod_{j=1}^{k-1} (N+j)}{(k-1)!}.$$

To calculate the number of possible convergent games, we must consider possibilities of odd and even parity. For odd parity, this is more possible, with a particular possibility for primes. In general for primes, if $N_{S_1} \equiv N_{S_2} \equiv \dots \equiv N_{S_k} \pmod{k}$ then the game will converge. Similarly, if $N_{S_1} \not\equiv N_{S_2} \not\equiv \dots \not\equiv N_{S_k} \pmod{k}$ then the game will not converge. From here, the invariant property can be utilised in a similar fashion to before to find the number of non-convergent games. Hence we find the number of convergent games. For even parity, we have no guarantee of these facts, so convergence is much rarer. This is as the invariant property here is not unique, so it becomes harder to find a possible convergent game.

Code

The code for this assignment can be found on my GitHub page, under a name relating to the Project Assignment:

<https://github.com/Bqrnie/Assignments>